Differentiation and coordination of gyral–sulcal structural networks reveal lifespan cortical reconfiguration associated with executive function

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Abstract:
The brain networks exhibit complex trajectories throughout life. Their reorganizations follow the differentiation and coordination processes of structural substrates. However, how these processes differentiate wiring resources and coordinate substrates across the lifespan remains remain. Here, we used a folding-informed framework to characterize differentiation and coordination in spatially embedded wiring topology using multimodal data from1687 healthy participants aged 8-89 years. For each topological feature, we defined a differentiation index (DI) to quantify the relative redistribution of wiring and topological profiles between gyral and sulcal subnetworks, and a synergy index (SI) to quantify relatively cross-type gyral–sulcal coordination. Using factor analysis, we identified four DI/SI axes that captured two major organizational principles: integration–segregation balance and cost–efficiency trade-off. Development was characterized by movement toward a relatively intermediate differentiation balance and the more efficient coordination between gyral and sulci. Young adulthood extended this pattern with weaker fine-tuning. Midlife emerged as a transition window in which gyri and sulci renegotiate wiring distributions and their coordination. Aging moved away from the intermediate differentiation balance and showed reduced cross-type coordination. Finally, early-life executive function (EF) improvement was mainly aligned with gyral relative stronger wiring differentiation, whereas later-life EF decline was linked to weaker cross-type wiring coordination. These findings highlight gyral–sulcal differentiation and coordination as key folding-informed principles of structural connectome reorganization, associated with cognitive performance.

Introduction:

Brain networks undergo continuous and nonlinear reorganization across development and aging (Liu et al., 2017; Mousley et al., 2025; Sun et al., 2025). Their topology, defined by the wiring diagram and associated functional interactions, provides a structural basis for accommodating changing cognitive demands and maintaining resilience during lifespan (Oberman & Pascual-Leone, 2013; Zamani Esfahlani et al., 2022; Zhao et al., 2015; Zuo et al., 2017). These topological reorganizations can be conceptualized through two complementary processes at the level of structural substrates: differentiation and coordination. At the cellular level, differentiation (or, reciprocally, dedifferentiation) has often been used to describe neural resource allocation, negotiating fidelity and precision of neural representations (Koen & Rugg, 2019). At the network level, dedifferentiation is extended into functional reorganization which may recruit compensatory neural resource to preserve cognition performance during aging, as proposed by the scaffolding theory (Deery et al., 2023; Reuter-Lorenz & Park, 2014). This dedifferentiation may manifest as a less segregated and less efficient network, related to the decrements of functional specialization (Deery et al., 2023; Naik et al., 2017). Likewise, in the structural connectome, we conceptualize differentiation as the reallocation of wiring cost profiles and redistribution of communication topology during lifespan. Such process may endow the substrates with topological profiles changes, supporting the functional specificity and cognitive demand (Deery et al., 2023; May, 2011; Reuter-Lorenz & Cappell, 2008; Schneider-Garces et al., 2010; Suárez et al., 2020). In contrast, coordination refers to the interaction among these differentiated substrates, optimizing the balance between neuronal resource cost and communication efficiency and supporting complex cognition processes (Bullmore & Sporns, 2012; Cohen & D'Esposito, 2016; Sporns, 2017; Wang et al., 2021). Together, the brain achieves the global integrative coordination of differentiated subsystems, revealing the ubiquitous integration–segregation balance under parsimoniously cost constraints (Bullmore & Sporns, 2012; Liao et al., 2017; Rubinov & Sporns, 2010). However, how differentiation and coordination reorganize across the lifespan remains insufficiently understood.

Previous studies have made pioneering progress in characterizing cortical differentiation and coordination through integration and segregation measures across lifespan. During development and young adulthood, within-module connections strengthen and structural modules may be increasingly differentiated, as reflected by the age-related increases and differentially distributed patterns in modular segregation (Baum et al., 2017; Deery et al., 2023). Although between-module connections weaken, costly edges of hubs strengthen and global efficiency increases, suggesting more efficient network coordination and partly mediating the executive improvement (Baum et al., 2017). In late adulthood, the within-system connections and segregation decrease, together with increased between-system connections and global efficiency (Chan et al., 2014). These patterns may indicate the less differentiated systems and higher cross-system coordination to optimize efficiency in response to cognitive decline (Chan et al., 2014; Mousley et al., 2025; Pedersen et al., 2021). However, existing approaches have mainly characterized absolute changes in structural wiring patterns. It remains unclear how differentiated substrates redistribute wiring resources and communication roles while remaining coordinated.

Gyri and sulci provide a folding-informed system to study differentiation and coordination across lifespan. First, they constitute two fundamental units of cortical folding, differentiating in genetic, cellular, structural, and functional patterns (Hilgetag & Barbas, 2005; Jiang et al., 2021; Llinares-Benadero & Borrell, 2019; Snyder et al., 2024; Xu et al., 2010). These differentiations indicate that gyri and sulci may occupy distinct positions in allocation of wiring diagram and integrative responsibility: gyral backbone and sulcal circuits. Specifically, gyri tend to be structural and functional hubs with denser and longer connections, whereas most sulci involved in local processing with only a small subset of core sulci contributing to the efficient global integration (He et al., 2022; Kruggel & Solodkin, 2023; Zhang et al., 2020). Importantly, they are also spatially interdigitated folding compartments to support the efficient coordination: sulci are embedded between surrounding gyri, and gyral crowns are connected with neighboring gyral and sulcal territories (Chen et al., 2017; Fernández-Pena et al., 2023). In this case, gyral-sulcal connections form cross-type bridge pathways, providing low-cost spatial routes through which sulcal circuits interact with the gyral backbone (Klyachko & Stevens, 2003; Striedter et al., 2015). Thus, the naturally differentiation and spatial embedding of folding make them well suited for studying topological differentiation and coordination. Second, folding provides a relatively stable anatomical scaffold while preserving individual variability across lifespan. Specifically, primary folds are highly conserved across individuals and species, the secondary is less conserved but more variable, and higher-order folds represent the greatest interindividual variability (Del-Valle-Anton & Borrell, 2022). Such individual-specific patterns serve as cortical "fingerprints" that are established at birth and close to the life-long configuration (Cachia et al., 2016; Snyder et al., 2024). Thus, this stable-but-individualized patterns reduce reliance on group-average templates or data-driven parcels and allows to focus on relative topological redistributions. Together, the folding provides an anatomically grounded system for characterizing structural differentiation and coordination across lifespan.

Here, we used the gyral–sulcal structural connectome as a folding-informed system to investigate how differentiation and coordination reorganize with age. Using 1687 healthy participants age 8–88 years from the Human Connectome Project (HCP) development, young adults and aging cohorts we constructed individual structural connectivity density and length matrix. We then defined two complementary classes of features from the graph topology. Differentiation index (DI) quantified relative redistribution of wiring cost and topological profiles between gyral and sulcal substrates. Additionally, synergy index (SI), quantified how gyral and sulcal substrates remain coordinated through cross-type gyral–sulcal connections. We then modeled nonlinear lifespan trajectories of these DI/SI features, identified respectively low-dimensional latent axes using factor analysis based on subject data, and projected trajectory curve into these axes to exam migrated patterns across development, adulthood, and aging. Finally, we examined whether the resulting DI/SI latent dimensions were associated with executive function.

Methods

Study participants

We used structural, diffusion MRI, and cognitive data from the Human Connectome Project Lifespan datasets, including the HCP Development dataset (HCP-D), the HCP Young Adult dataset (HCP-YA), and the HCP Aging dataset (HCP-A). Because the number of participants younger than 8 years or older than 89 years was relatively sparse, our analyses were restricted to participants aged 8–89 years.

Because HCP-YA included siblings, we randomly selected one participant from each family. During this selection procedure, we prioritized individuals retained in our previous analyses and individuals with complete cognitive measures, while also maintaining an approximately balanced sex distribution. After this procedure, the final HCP-YA sample included 419 participants, with 210 males and 209 females.

The final sample included 1687 healthy participants across the three datasets. Specifically, HCP-D included 592 participants (273 males and 319 females; age range: 8–22 years; mean age: 14.63 ± 3.90 years). HCP-YA included 419 participants (210 males and 209 females; age range: 22–37 years; mean age: 28.63 ± 3.64 years). HCP-A included 676 participants (295 males and 381 females; age range: 36–89 years; mean age: 59.45 ± 14.74 years). Overall, the sample included 778 males (46.12%) and 909 females (53.88%). Sex distribution was relatively balanced in HCP-YA, whereas HCP-D and HCP-A included slightly higher proportions of female participants.

All participants provided written informed consent. For participants younger than 18 years, permission was acquired from a parent or legal guardian. HCP-D participants were scanned at Boston, Los Angeles, Minneapolis, and St. Louis, and information were found from the HCP-D 2.0 Release. HCP-YA participants from the HCP S1200 Release were scanned at Washington University in St. Louis. HCP-A participants from the HCP-A 2.0 Release were scanned at Washington University in St. Louis, University of Minnesota, Massachusetts General Hospital, and University of California Los Angeles.

Covariates used in subsequent analyses included sex, estimated total intracranial volume (eTIV), and handedness. Individual-level executive function (EF) was assessed using four NIH Toolbox measures: Flanker Inhibitory Control and Attention (Flanker), Dimensional Change Card Sort (DCCS), List Sorting Working Memory (LSWM), and Pattern Comparison Processing Speed (PCPS).

Based on the age coverage of dataset, we divided all participants into four broad lifespan epochs: development ((8, 22) years), young adulthood ([22, 35] years), midlife ((35, 65] years), and aging ((65, 89) years). In addition, each dataset was further divided into five age bins for visualization and stratified analyses. The age and sex distributions of the final sample are shown in Fig. 1a.

MRI acquisition and preprocessing

Although all participants were scanned on 3T MRI scanners, the specific scanner hardware and parameter settings differed slightly across HCP Lifespan datasets. These differences reflected both challenges and scientific priorities when imaging younger and older populations. In general, these acquisition protocols were designed to be similar but differed.

For HCP-YA, data were acquired on a customized Siemens 3T Connectome Skyra scanner equipped with a 100 mT/m gradient coil. T1-weighted structural data were acquired via a single-echo sequence with echo time (TE) = 2.1 ms, repetition time (TR) = 2400 ms, field of view (FOV) = 224×224 mm^2, and 0.7 mm isotropic voxels. Diffusion MRI data were acquired with multiple b-values of 1000, 2000, and 3000 s/mm^2, with diffusion TR = 3.23 s and 1.5 mm voxels.

For HCP-D and HCP-A, data were acquired on Siemens 3T Prisma scanners equipped with an 80 mT/m gradient coil. In T1-weighted structural images, a multi-echo sequence was used with TE = 1.8/3.6/5.4/7.2 ms, TR = 2500 ms, FOV = $256 \times 240 \times 166$ mm^3, and 0.8 mm isotropic voxels. For Diffusion MRI, data were obtained with b-values of 1500 and 3000 s/mm^2, with diffusion TR = 5.52 s and 1.25 mm voxels. Detailed acquisition parameters are available in the official HCP documentation.

All neuroimaging data were preprocessed using the HCP minimal preprocessing pipelines. Briefly, for Structural MRI data, native structural spaces were generated, T1-weighted and T2-weighted images were aligned, and structural images were registered to MNI space. Brain tissue segmentation was performed, and white and pial cortical surfaces were reconstructed. Surface and volume data were then generated, followed by MSM-Sulc surface registration and downsampling. Diffusion MRI data were preprocessed using the structural outputs and diffusion scans, including image intensity normalization, correction for EPI distortions, eddy-current-induced distortions, head motion, and gradient-nonlinearity distortions, followed by registration between diffusion and the structural data.

Probabilistic tractography and structural connectivity density and length matrix

Structural connectivity was reconstructed using probabilistic tractography. For HCP-YA, we downloaded the minimally preprocessed 3T diffusion MRI data and the corresponding BedpostX outputs. For HCP-D and HCP-A without preprocessed diffusion parts, the HCP minimal preprocessing pipeline was first applied to the raw diffusion MRI data, followed by probabilistic tractography.

Diffusion MRI processing consisted of three main steps: fiber orientation density function estimation, probabilistic tractography, and network reconstruction. The first part was estimated using FSL's BedpostX. For HCP-YA, precomputed BedpostX outputs were used. Probabilistic tractography was then performed using FSL's probtrackx2.

After that, we used waytotal-normalized tractography outputs to reduce inter-individual differences in the total number of generated streamlines. Specifically, the dense values were divided by the corresponding waytotal value, converting streamline counts into proportional estimates. A log transformation was then applied to reduce the distance-related bias, particularly the under-representation of long-range projections.

These processed dense connectomes were then mapped to the individualized Destrieux gyral–sulcal atlas provided by HCP, rather than to a standard template atlas. This yielded subject-specific structural connectivity matrices between gyral and sulcal cortical regions.

For the streamline count matrix, we further computed streamline count density to account for differences in parcel size. Because individualized Destrieux gyral and sulcal regions vary in size, larger parcels may have a higher probability of receiving streamlines. For each pair of regions, the log-transformed streamline count was divided by the average relative size of the two regions, where each parcel size was normalized by the total cortical volume. This produced a size-corrected structural connectivity density matrix.

For the streamline length matrix, we used within-subject percentile ranks of connection length ranging from 0 to 1. This transformation reduced the impact of between-subject differences in absolute length scale while preserving the relative organization of short- and long-range connections within each individual.

Gyral-sulcal parcellations and definitions of topological organization

From our previous work, the original 148 Destrieux regions were binarized into gyral and sulcal labels, yielding 34 gyral and 34 sulcal regions in each hemisphere. Based on these labels, we partitioned each subject's structural connectivity matrix into two subnetwork types: within-module (gyral–gyral: G–G; sulcal–sulcal: S–S) and between-module connections (gyral–sulcal: G–S).

Because the overall connectivity strength differed across subnetwork types, with G–G connections generally being strongest, S–S connections weakest, and G–S connections intermediate, we applied a fixed-density 10%-10%-10% thresholding scheme separately to the three subnetworks. Specifically, for each subject, the top 10% of G–G edges, top 10% of S–S edges, and top 10% of G–S edges were retained. This scheme ensured comparable edge densities across within-module and between-module subnetworks, allowing topological measures to be compared without being dominated by differences in edge counts. The resulting subject-specific binary masks were then used to compute structural topological measures from the connectivity density and length matrices.

From the perspective of five organizational mechanisms, we defined nine topological features in five mechanistic families (scale/cost, dispersion, local redundancy, bottleneck, and concentration).

In the scale/cost group, we used connection weight (*weight*) and length (*len*) to capture the overall strength and wiring distance of structural connections. *weight* was defined as the mean weight of retained structural connections. *len* was defined as the average within-subject percentile-ranked streamline length across retained edges. Higher *weight* or *len* indicates relatively stronger or long-range connectivity.

In the dispersion group, we used participation coefficient (*PC*) and global efficiency (*Eg*) to capture distributed integration. A higher *PC* value (corrected by $PC/(1-1/K)$, where K is the number of communities) indicates that connections are more evenly distributed across communities. The higher *Eg* shows more efficient communication. For gyral–sulcal connections, *Eg* was computed between gyral and sulcal nodes, reflecting the efficiency of cross-type communication.

In the local redundancy group, we used local efficiency (*Eloc*) and clustering coefficient (*CC*) to capture local fault tolerance. *Eloc* measured how efficiently the neighbors of a node remained connected with each other. *CC* quantified the extent to which a node's neighbors formed locally closed or clustered patterns. These two measures were adapted to evaluate whether a node's opposite-type neighbors formed locally efficient or clustered anchoring structures for cross-type connections.

In the bottleneck group, we used the betweenness ratio (*BE_ratio*) to capture dominant dependence. Betweenness (*BE*) was first calculated. The *BE_ratio* was then defined as the ratio between the gyral and sulcal mean betweenness values. Values greater (or less) than one indicate greater reliance on gyral (or sulcal) intermediates along nodal communication paths.

In the concentration group, we used edge concentration (*Conc_edge*) and long-range concentration (*Conc_len*) to measure the specificity of community-pair connectivity (corrected by $(C-1/n)/(1-1/n)$, where n is the number of efficient edges among communities). *Conc_edge* measured whether retained edges were broadly distributed (lower value) across many communities or concentrated (higher value) within a small number of community-pair connections. *Conc_len* extended this idea to connection length property. Thus, Higher *Conc_edge* (*Conc_len*) values indicate more selective and specific (longer) community-pair organization. Both values are corrected by the number of communities.

Together, these nine features provided complementary descriptions of gyral–sulcal structural organization. Node-level measures were averaged across gyral or sulcal nodes to obtain one global value.

Definition of differentiation (DI) and synergy (SI) indices

To characterize gyral–sulcal structural organization, we defined two complementary classes of indices: the Differentiation Index (DI) and the Synergy Index (SI). DI was designed to capture how gyri and sulci differ in their topological organization, whereas SI was designed to quantify how gyri and sulci coordinate through cross-type structural connections. In brief, DI asks how different gyri and sulci are, whereas SI asks how they work together.

For a given topological feature X, the DI was defined as the relative difference between the gyral and sulcal values:

$$DI_X = \frac{X_{GG} - X_{SS}}{(X_{GG} + X_{SS})/2}$$

where X_{GG} and X_{SS} denote the feature values computed for gyral and sulcal subnetworks, respectively. A positive value indicates that the feature is stronger in gyri than in sulci, vice versa. Larger absolute DI (or closer to zero) values indicate stronger differentiation (or relative dedifferentiation) between gyri and sulci.

Similarly, SI was defined by comparing the cross-type gyral–sulcal value with the average of within-type gyral–gyral and sulcal–sulcal values:

$$SI_X = \frac{X_{GS}}{(X_{GG} + X_{SS})/2} - 1 = \frac{X_{GS} - (X_{GG} + X_{SS})/2}{(X_{GG} + X_{SS})/2}$$

where X_{GS} denotes the feature value computed for gyral–sulcal connections. Subtracting 1 centers the index around zero. Thus, positive (negative) SI values indicate that cross-type gyral–sulcal organization is stronger (weaker) than the average within-type organization, providing a useful reference for whether cross-type organization is relatively enhanced or reduced.

Notably, we normalized DI and SI by the mean within-type value, rather than by the whole-brain value, because our goal was to quantify gyral–sulcal differentiation and cross-type coordination relative to the within-type organizational baseline. For DI, this denominator provides a symmetric relative contrast between gyral and sulcal subnetworks without incorporating cross-type gyral–sulcal connections. For SI, the same denominator serves as a reference for within-type organization, allowing positive and negative values to indicate whether cross-type gyral–sulcal organization is enhanced or reduced relative to the average within-type organization. In contrast, normalization by a whole-brain value would mix within-type and cross-type contributions and may confound the indices with global network strength or age-related whole-brain changes.

Importantly, the sign of DI has a direct interpretation because it reflects a relative contrast between gyral and sulcal values for the same within-module definition. In contrast, the sign of SI should be interpreted more cautiously since cross-module features may be defined in a way that is analogous but not identical to their within-type counterparts. Therefore, SI was interpreted primarily as a relative measure of cross-type coordination rather than as an absolute directional contrast. Finally, 9 topological features were used to compute both the DI and SI, yielding 18 measures for each participant.

Lifespan trajectory modeling and stratified bootstrap estimation

To model lifespan trajectories of gyral–sulcal structural organization, we used generalized additive models for location, scale, and shape (GAMLSS). GAMLSS was applied to each DI/SI feature to estimate age-related changes from 8 to 89 years. Following a similar normative modeling strategy, age was modeled using B-spline smooth terms, and sex and estimated total intracranial volume (eTIV) were included as covariates. Dataset-related effects were harmonized using the *neuroHarmonize* toolbox before lifespan curve estimation. We did not include dataset/site as an explicit covariate in the GAMLSS models after harmonization.

For each feature, we fitted three candidate models with different degrees of freedom (df=3,4,5) for the age spline term. The model with the best the Bayesian information criterion (BIC) was selected, while convergence was also checked. During fitting, the Johnson's SU (JSU) distribution was used. We obtained seven centile curves (1st, 5th, 25th, 50th, 75th, 95th, and 99th percentiles) from the best model. The 50th centile was used as the main lifespan trajectory.

In addition, we also extracted covariate-adjusted subject-level feature values from the fitted GAMLSS models. Specifically, after model fitting, the effects of eTIV and sex covariates were subtracted. These residuals were used for downstream analyses, including feature–age association, latent factor analysis and factor-cognition association.

To estimate uncertainty and reduce the influence of extreme observations, we performed stratified bootstrap resampling. Each dataset was divided into five age bins. The bin boundaries were 8, 11, 14, 17, 20, 22, 25, 27, 30, 33, 35, 45, 55, 65, 75, and 89 years. Within each age bin, participants were sampled with replacement, and the GAMLSS model was refitted. This procedure was repeated 1,000 times. This strategy was used because simple resampling could lead to insufficient age coverage in some portions, especially in HCP-A, where the age range is broad (36-89 years) but the number of participants (N=676) is relatively limited.

After growth curves were estimated, GAM (generalized additive models) smoothing was applied to reduce local irregularities. Here, a univariate LinearGAM with a spline term for age and 10 splines was fitted, and the predicted values were used as the smoothed trajectory.

Factor analysis and state-space trajectory

We hypothesized that these DI/SI features could be represented by a lower-dimensional latent structure (differentiation and synergy). To identify these latent dimensions, we applied factor analysis (FA) to covariate-adjusted subject-level DI and SI features. DI and SI features were analyzed separately to preserve their conceptual distinction: DI captured gyral–sulcal differentiation, whereas SI captured cross-type gyral–sulcal coordination. Here, FA decomposed the observed feature covariance into shared variance, reflecting a latent organizational system.

Based on the loading structure and interpretability, we both retained two latent axes for DI and SI features: DI-FA1, DI-FA2, SI-FA1, and SI-FA2. Then, the modeled lifespan curves were then transformed to obtain age-varying latent scores, providing a low-dimensional representation of gyral–sulcal differentiation and coordination reconfiguration.

Executive function modeling, mediation, and age-dependent coupling

To examine the behavioral relevance of latent gyral–sulcal organization, we constructed an executive function (EF) latent factor using cognitive measures from the NIH Toolbox. Four cognitive tasks were included: Flanker, DCCS, LSWM, and PCPS. We then estimated a common EF factor within each dataset.

We first tested whether latent DI/SI dimensions mediated the relationship between age and executive function using a classical mediation framework. The four latent brain factors, DI-FA1, DI-FA2, SI-FA1, and SI-FA2, were included as parallel mediators. The sex, handedness, eTIV and dataset were used as covariates.

$$M_k = \alpha_{0k} + a_k \cdot Age + covariates + \epsilon_{Mk}$$

$$EF = \beta_0 + c' \cdot Age + \sum_k b_k \cdot M_k + covariates + \epsilon_{EF}$$

$$Indirect\ effect_k = a_k \times b_k = \frac{dM_k}{dAge} \times \frac{dEF}{dM_k}$$

where a represents the association between age and the latent DI/SI factor, b represents the association between the latent DI/SI factor and EF after controlling for age, and c represents the direct age effect on EF after accounting for the mediator.

However, classical mediation provides only a global estimate and does not describe how the age–brain–cognition pathway varies across the lifespan. Therefore, we further defined a semi-parametric local mediation-like index (*semi_LM*) to quantify age-dependent pathway contribution.

$$EF = \beta_0 + \beta_{age}Age + \sum_k \beta_{M_k} M_k + \sum_k \beta_{age\times M_k}(Age \times M_k) + C\gamma + \epsilon$$

where C included dataset, sex, handedness, and eTIV covariates.

$$\frac{dEF}{dM}(age) \approx \frac{EF(age, M + \Delta) - EF(age, M - \Delta)}{2\Delta}$$

$$semi\_LM(age) = \frac{dM_{GAMLSS}}{dAge}(age) \times \frac{dEF_{linear\ interaction}}{dM}(age)$$

In this *semi_LM* equation, the first term was derived from the fitted GAMLSS-based latent trajectory by computing its first derivative. The second term was estimated from a linear interaction model. This positive (negative) *semi_LM* index was used as a descriptive measure of whether age-related changes in latent gyral–sulcal organization were locally aligned with increases (decreases) in executive function.

For uncertainty estimation, this *semi_LM* index was computed within each bootstrap iteration. Specifically, subject-level data were resampled, GAMLSS lifespan curves were refitted, the resulting DI/SI trajectories were projected into the fixed reference FA space defined from the original subject data, and the linear interaction were also refitted. The product of these two terms was then computed.

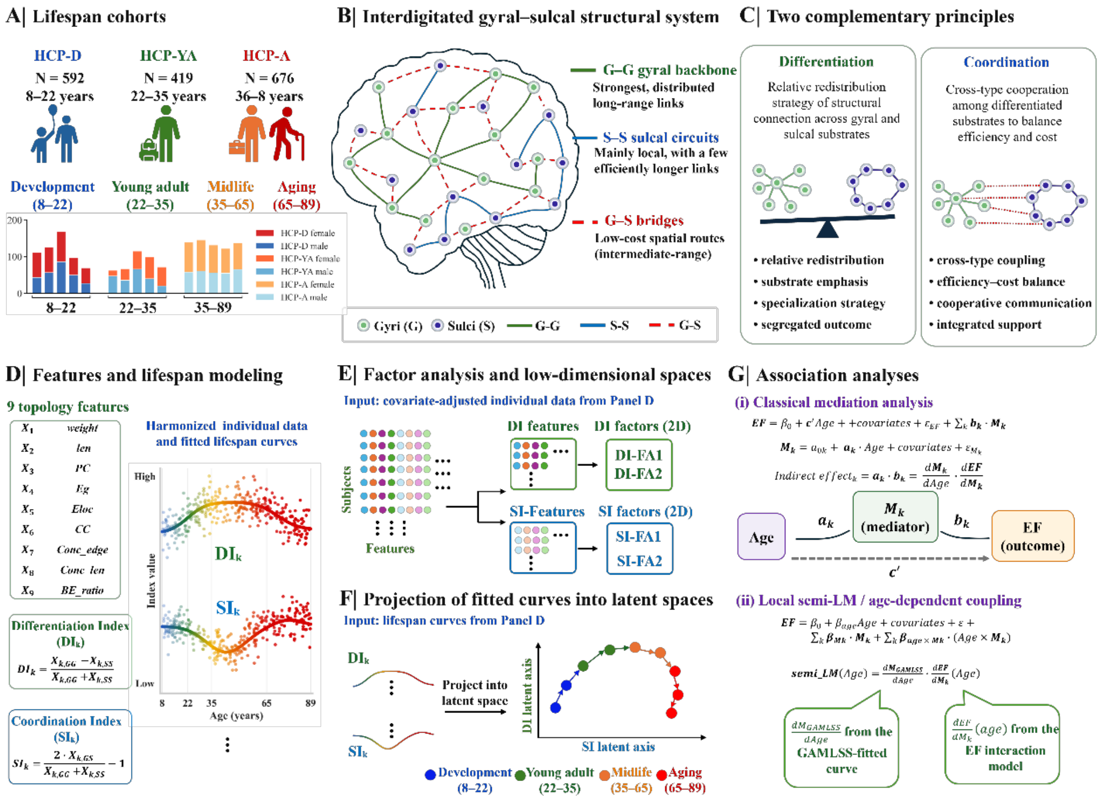

Fig. 1. Folding-informed framework for quantifying gyral–sulcal differentiation and coordination. Individual structural connectomes were constructed from multimodal imaging data and mapped onto a personalized gyral–sulcal atlas. Cortical nodes were labeled as gyral (G) or sulcal (S), and structural connections were decomposed into three subnetwork types: gyral–gyral (G–G), sulcal–sulcal (S–S), and gyral–sulcal (G–S) connections. For each topological feature (X), we computed feature values from the G–G, S–S, and G–S subnetworks. The differentiation index (DI), (G-G – S-S) / (G-G + S-S), quantified the relative redistribution of wiring and topological profiles between gyral and sulcal within-type subnetworks. Positive and negative DI values indicate gyral and sulcal dominance, respectively, whereas larger absolute DI values indicate stronger gyral–sulcal differentiation. The synergy index, 2*G-S/(G-G + S-S) - 1, quantified cross-type gyral–sulcal coordination relative to the within-type baseline. Positive SI values indicate relatively stronger G–S coordination, whereas negative SI values indicate weaker cross-type coordination. DI and SI features were further embedded into low-dimensional latent spaces using factor analysis, yielding 4 axes related to integration–segregation balance and wiring allocation/cost-efficiency organization. These latent axes were then used to characterize nonlinear lifespan trajectories and their associations with executive function.

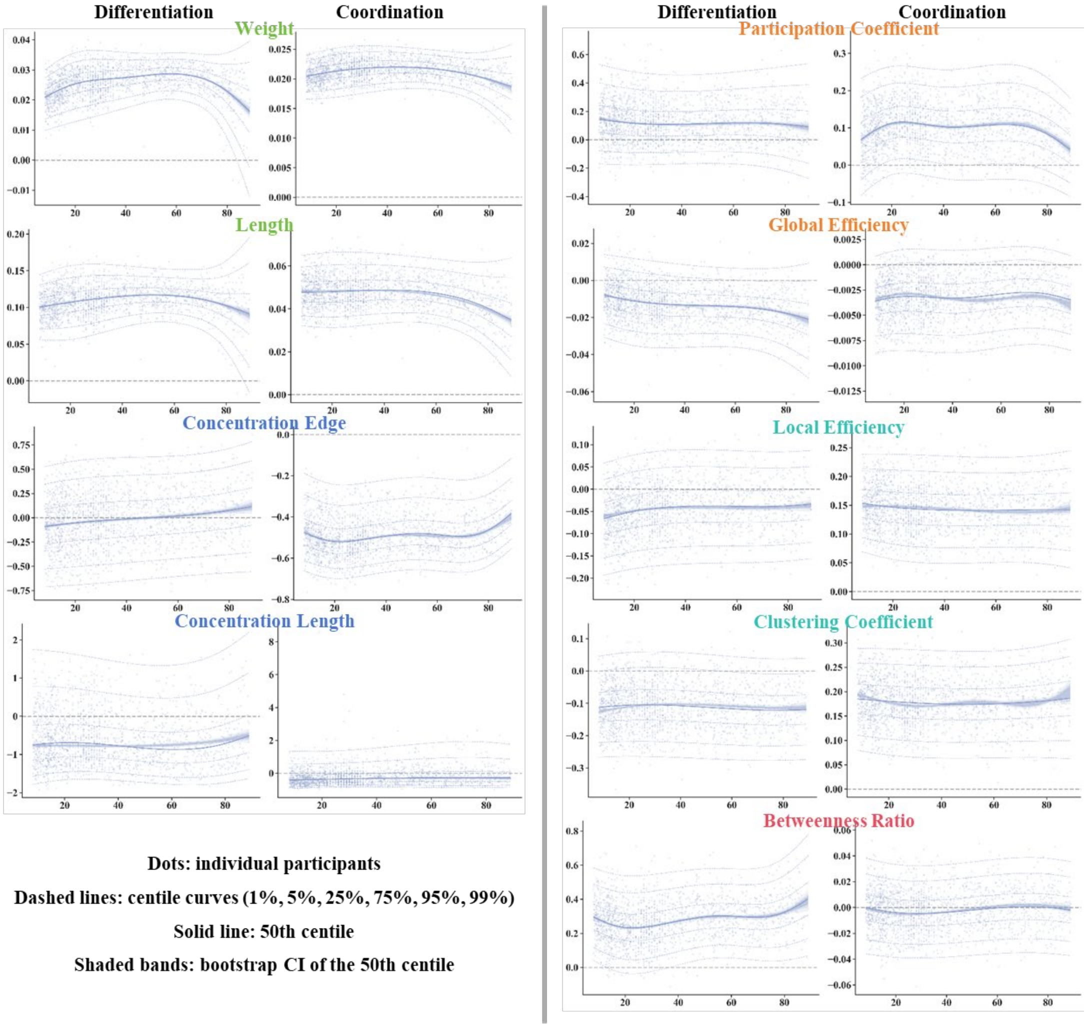

Fig. 2. Lifespan curves of differentiation and coordination in folding-informed systems. Lifespan trajectories were estimated using GAMLSS models. The x-axis represents age, and the y-axis represents the feature values. Dots indicate individual participants after adjusting for eTIV and sex. The fitted centile curves are shown for the 1st, 5th, 25th, 50th, 75th, 95th, and 99th percentiles. Dashed lines represent the non-median centiles, whereas the solid line represents the median trajectory. Shaded bands indicate the confidence interval of the 50th centile, estimated from 1000 Bootstrap resampling and model refitting procedures. Abbreviation: GAMLSS (generalized additive models for location, scale, and shape), eTIV (estimated total intracranial volume)

Results:

Growth curves of topological differentiation and coordination

We used GAMLSS to model lifespan trajectories of gyral–sulcal structural features from 8 to 89 years. These features quantified two complementary aspects of folding-informed network organization: differentiation between gyral and sulcal subnetworks (DI features), as well as coordination across gyral–sulcal connections (SI). The modeled

DI and SI trajectories are shown in Fig. 2, whereas the original G–G, S–S, and G–S subnetwork trajectories are shown in Fig. S1.

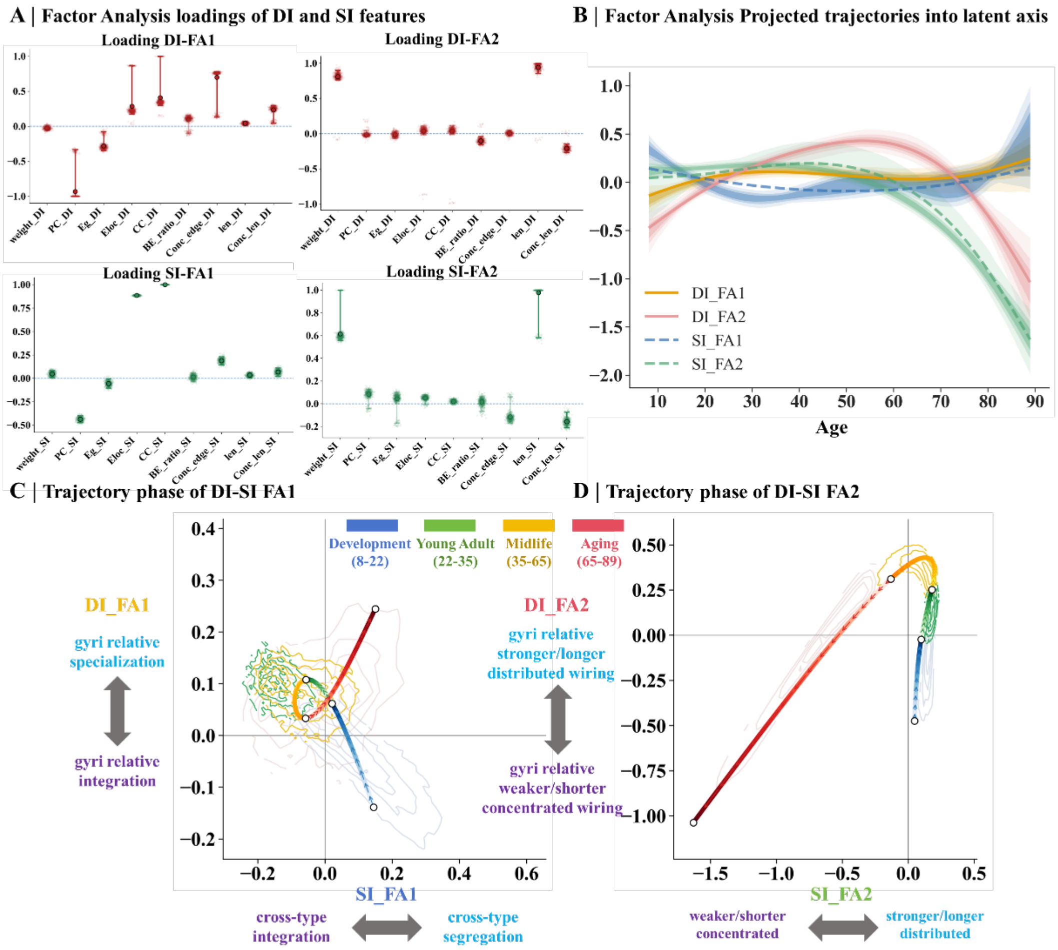

Fig. 3. Factor loadings and lifespan trajectories of DI/SI latent organizational axes. **(A)** Factor loading structure of 4 differentiation- and coordination-related latent axes derived from individual-level features adjusted by model. The top two panels show DI-FA1 and DI-FA2, while bottle panels show SI-FA1 and SI-FA2. Each x-axis label represents a topological feature, and the y-axis represents the factor loading. Semi-transparent points represent the bootstrap distribution of loadings, solid points with black edges indicate the original loading, and vertical error bars indicate the 95% bootstrap confidence interval. Features with confidence intervals crossing zero were shown with lower opacity. (B) Lifespan trajectories of the four latent organizational axes after projecting age-varying curves into the individual-defined DI/SI latent space. The x-axis represents age in years, and the y-axis represents the low-dimensional factor score. Solid lines indicate DI factor trajectories, and dashed lines indicate SI factor trajectories. Orange, pink, blue, and green represent DI-FA1, DI-FA2, SI-FA1, and SI-FA2, respectively. Shaded bands represent bootstrap uncertainty of the fitted trajectories. Darker shading indicates greater concentration of bootstrap trajectories and therefore higher trajectory stability at that age. (C)(D) Phase-space representations of

lifespan trajectories of projected curves in the DI/SI latent space. The trajectories summarize age-related transitions across developmental, young-adult, midlife, and aging epochs. Abbreviation: FA (Factor analysis)

Four latent axes and their stability of differentiation and coordination

We next asked whether gyral-sulcal differentiation and coordination could be represented by lower-dimensional latent organizational axes. To this end, we applied factor analysis (FA) separately to these DI and SI features, allowing shared variance across topological measures to define latent dimensions. This analysis consistently identified two DI and two SI axes, shown in Fig. 2A.

DI-FA1 was characterized by opposite loadings between integration and segregation features. The positive end was mainly defined by Conc_edge_DI (0.76, 95% CI [0.13, 0.78]), CC_DI (0.34, 95% CI [0.30, 1.00]), Conc_len_DI (0.25, 95% CI [0.04, 0.30]), Eloc_DI (0.22, 95% CI [0.17, 0.87]), and len_DI (0.04, 95% CI [0.01, 0.07]). In contrast, the negative was dominated by PC_DI (-1.00, 95% CI [-1.00, -0.33]) and Eg_DI (-0.29, 95% CI [-0.34, -0.08]). Thus, we interpreted this factor as an integration–segregation differentiation axis, with the negative-to-positive direction reflecting a transition from gyral-relative integration toward gyral-relative specialization.

DI-FA2 was primarily defined by wiring scale/cost allocation features. The positive end was dominated by len_DI (0.95, 95% CI [0.86, 1.00]) and weight_DI (0.82, 95% CI [0.76, 0.90]). The negative end was defined by Conc_len_DI (-0.22, 95% CI [-0.27, -0.15]) and BE_ratio_DI (-0.10, 95% CI [-0.16, -0.04]). We therefore interpreted DI-FA2 as a wiring allocation or cost differentiation axis. Negative to positive end reflects a shift from weakening gyral-relative backbone wiring toward stronger and strengthening gyral-relative distributed wiring patterns.

SI-FA1 also captured a integration–segregation contrast in gyral-sulcal coordination. The positive end was strongly defined by CC_SI (1.00, 95% CI [1.00, 1.00]) and Eloc_SI (0.89, 95% CI [0.87, 0.89]), with additional positive contributions from Conc_edge_SI (0.18, 95% CI [0.14, 0.23]), Conc_len_SI (0.07, 95% CI [0.02, 0.11]), weight_SI (0.06, 95% CI [0.01, 0.09]), and len_SI (0.03, 95% CI [0.01, 0.06]). In contrast, the negative end was defined by PC_SI (-0.43, 95% CI [-0.47, -0.40]) and Eg_SI (-0.05, 95% CI [-0.10, -0.01]). Thus, SI-FA1 was interpreted as a cross-type integration (negative) to segregation (positive) axis between gyri and sulci.

SI-FA2 reflected the wiring allocation and cost dimension of gyral–sulcal coordination. The positive end was mainly defined by len_SI (1.00, 95% CI [0.58, 1.00]) and weight_SI (0.60, 95% CI [0.56, 1.00]), indicating stronger and longer cross-type wiring. The negative end was defined by Conc_len_SI (-0.17, 95% CI [-0.21, -0.08]), indicating relatively more concentrated long-range backbone organization. We therefore interpreted SI-FA2 as a wiring allocation of coordination, ranging from weaker/shorter long-range concentrated to stronger/longer distributed cross-type wiring.

Together, these four latent axes suggest that DI/SI features form structured organizational dimensions rather than a collection of independent graph metrics. Across DI and SI, the first factors primarily captured an integration–segregation balance, whereas the second factors captured wiring allocation and cost-related organization. As a supplementary analysis, the joint FA solution, shown in Fig. S2, revealed comparable axes related to segregation–integration balance of DI and SI and strength/length scale, supporting the robustness of the main DI/SI factor structure.

Low-dimensional lifespan trajectories in DI/SI state space

Once latent spaces were defined from individual-level DI and SI features using factor analysis, the GAMLSS-derived lifespan curves were then projected into the same reference space. This procedure yielded four age-varying latent trajectories , providing a low-dimensional differentiation–coordination (DI-SI) state representation of gyral-sulcal reorganizations, shown in Fig. 3B & 3C.

The DI-SI FA1 phase plane primarily described integration-segregation differentiation balance and cross-type coordination, shown in Fig. 3C. Under the 80% CI criterion, shown in Fig. 4D, during development, DI-FA1 significantly increased, whereas SI-FA1 decreased. In the phase plane, this corresponds to an upward and leftward trajectory. Young adulthood continued the tail of this developmental pattern, with DI-FA1 increasing until approximately 31.5 years and SI-FA1 decreasing until approximately 28.5 years. This suggests a late maturation or fine-tuning stage. Midlife showed a different pattern. DI-FA1 decreased between approximately 48.4 and 58.3 years, while SI-FA1 transiently increased between approximately 41.6 and 51.1 years. In the phase plane, this suggests a local reversal or turning pattern, moving downward and slightly rightward. This midlife change may reflect a rebalancing of integration–segregation organization between substrate differentiation and cross-type coordination. In aging, DI-FA1 and SI-FA1 showed only brief significant intervals, with DI-FA1 increasing from 78.0 years and SI-FA1 increasing around 73.8–76.0 years. These changes were interpreted cautiously as late-life fluctuation or possible local compensation.

The DI-SI FA2 phase plane described wiring allocation and cost-related patterns, shown in Fig. 3D & 4D. During development, DI-FA2 significantly increased, whereas SI-FA2 did not show significant change. Thus, the phase trajectory mainly moved upward, suggesting that early wiring redistribution was primarily driven by differentiation-related wiring allocation. Young adulthood continued this upward movement. Midlife was the key transition. DI-FA2 continued to increase until approximately 51.1 years, whereas SI-FA2 began to significantly decrease from approximately 37.4 years onward. This indicates that the system moved upward and leftward. In aging, both DI-FA2 and SI-FA2 decreased significantly, with DI-FA2 decreasing from approximately 58.0 years onward and SI-FA2 continuing its decrease. In the phase plane, this corresponded to a downward and leftward trajectory, indicating a broad age-related decline or redistribution along the wiring allocation and cost-efficiency dimension.

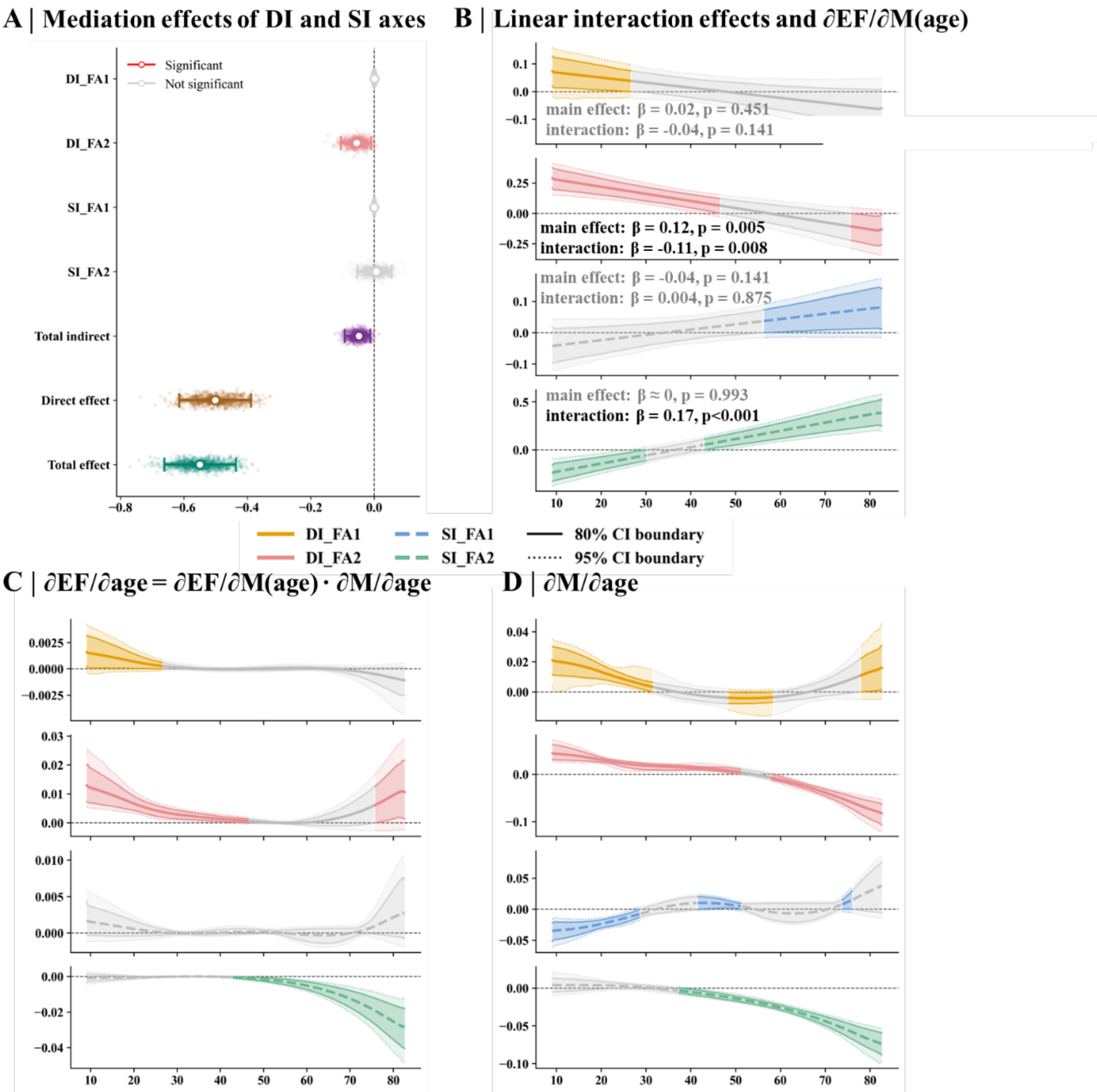

Fig. 4. Mediation, age-interaction, and lifespan derivative analyses linking DI/SI latent axes to executive function. (A) Parallel mediation analysis testing whether the association between age and EF was mediated by the four DI/SI latent axes, including DI_FA1, DI_FA2, SI_FA1, and SI_FA2. Colored points and intervals indicate significant effects, whereas gray points indicate non-significant effects. (B) Linear age-interaction models assessing whether the associations between DI/SI latent axes and EF varied with age. The annotated text reports the main effect and age-interaction effect. The y-axis represents the age-dependent EF association, ∂EF/∂M(age), where M

denotes the latent factor score. Lighter outer shading indicates the 95% bootstrap CI, and darker inner shading indicates the 80% bootstrap CI. Dotted and solid outlines denote the 95% and 80% CI boundaries, respectively. Curves are colored when the 80% CI does not cross zero and shown in gray otherwise. DI axes are shown as solid lines, and SI axes are shown as dashed lines. (C) EF-aligned age-related changes estimated using semi_LM measures: $\partial EF/\partial age = \partial EF/\partial M(age) \cdot \partial M/\partial age$. Positive values indicate age-related changes aligned with higher EF, whereas negative values indicate changes aligned with lower EF. Shading, CI boundaries, significance coloring, and line styles follow panel B. (D) First derivatives ($\partial M/\partial age$) of the fitted lifespan trajectories of the DI/SI latent axes. Positive values indicate age-related increases, whereas negative values indicate age-related decreases. Shading, CI boundaries, significance coloring, and line styles follow panel B. Abbreviation: CI (confidence interval), EF (executive function)

Age-dependent executive function (EF) coupling of DI/SI latent dynamics

We next examined whether these latent differentiation and coordination dimensions mediated the association between age and EF. In a classical parallel mediation model, DI-FA1, DI-FA2, SI-FA1, and SI-FA2 were entered simultaneously as mediators while controlling for dataset, sex, handedness, and eTIV. The total indirect effect was significantly negative (effect = -0.05, 95% CI [-0.09, -0.01]), explaining approximately 8.7% of the total age effect. Among the four mediators, DI-FA2 was the only factor showing a stable individual indirect effect, shown in Fig. 4A. Age was negatively associated with DI-FA2 ($a = -0.39$, $p < 0.001$), whereas DI-FA2 was positively associated with EF after controlling for age and the other mediators ($b = 0.14$, $p < 0.001$). This yielded a significant negative indirect effect ($a \times b = -0.06$, 95% CI [-0.11, -0.01]). In contrast, DI-FA1, SI-FA1, and SI-FA2 did not show stable individual indirect effects.

We next examined whether the latent axes were associated with EF in an age-dependent manner. We first fitted a linear interaction model with EF as the dependent variable and age interactions with DI-FA1, DI-FA2, SI-FA1, and SI-FA2, shown in Fig. 4B. The model explained a significant proportion of EF variance (adjusted $R^2 = 0.087$, $p < 0.001$). Age was negatively associated with EF ($\beta = -0.42$, $p < 0.001$). Among the four latent dimensions, DI-FA2 showed a positive main effect on EF ($\beta = 0.12$, $p = 0.005$), whereas its interaction with age was significantly negative ($\beta = -0.11$, $p = 0.008$). This indicates that the positive association between DI-FA2 and EF was stronger in early life and weakened with age. In contrast, SI-FA2 showed no significant main effect ($\beta \approx 0$, $p = 0.993$), but its interaction with age was significantly positive ($\beta = 0.17$, $p<0.001$), indicating that the association between SI-FA2 and EF became stronger at older ages. DI-FA1 and SI-FA1 showed no significant main effects or age interactions. Specifically, DI-FA1 was not significantly associated with EF as a main effect ($\beta = 0.02$, $p = 0.451$), and its interaction with age was also not significant ($\beta = -0.04$, $p = 0.141$). Similarly, SI-FA1 showed no significant main effect on EF ($\beta = 0.004$, $p = 0.875$), and its age interaction was not significant ($\beta = 0.03$, $p = 0.159$).

To further localize the age intervals in which DI/SI latent dynamics were strongly aligned with EF, we used a semi-local mediation-like framework based on modeled age-related changes with its age-dependent interaction, referred to as semi_LM, shown in Fig. 4C. A positive (negative) value indicates that age-related changes in a DI/SI latent factor are locally aligned with the direction of higher (lower) EF. Using the 80% bootstrap CI criterion, DI-FA1 showed a positive semi_LM contribution from 9.1 to 26.5 years. This interval overlapped with both positive age-related change in DI-FA1 and positive DI-FA1–EF coupling. DI-FA2 showed positive semi_LM contributions from 9.1 to 46.5 years and again from 75.7 years. During the earlier interval, DI-FA2 increased with age and showed positive EF coupling. During the late-life interval, DI-FA2 decreased with age and showed negative EF coupling, also yielding a positive local contribution. SI-FA1 did not show a stable semi_LM contribution. In contrast, SI-FA2 showed a negative semi_LM contribution from 43.0 years, characterized by a sustained decrease in SI-FA2 together with positive SI-FA2–EF coupling.

Discussion:

In this study, we used structural connectivity density and length matrices to characterize nonlinear and low-dimensional reorganization of gyral–sulcal differentiation and coordination from 8 to 89 years. By defining DI and SI across topological features, we showed that structural organization can be embedded into four stable latent axes. Specifically, DI-FA1 and SI-FA1 captured integration–segregation balance, with positive loadings mainly from concentration, clustering, and local efficiency features, and negative loadings from participation coefficient and global efficiency. DI-FA2 and SI-FA2 captured wiring allocation and cost, with positive loadings from connection length and strength and negative loadings from long-range concentration. Projecting GAMLSS-derived lifespan curves into this latent DI/SI space revealed stage-specific trajectories, with relatively balanced differentiation and increased coordination during development, selective fine-tuning in young adulthood, transitional rebalancing in midlife, and relatively unbalanced differentiation and reduced coordination in aging. Importantly, these structural trajectories were behaviorally relevant in an age-dependent manner: EF improvement in early life was mainly aligned with differentiation-related wiring organization, whereas EF decline from midlife to aging was linked to weaker cross-type wiring coordination. Together, our findings suggest that lifespan cortical reconfiguration and its cognitive relevance can be understood through folding-informed low-dimensional patterns of gyral–sulcal differentiation and coordination.

A multi-level gyral–sulcal integration framework for differentiation and coordination

The four differentiation–coordination axes can be further understood within a spatially embedded, multi-level gyral–sulcal integration framework. DI factors capture the population-referenced conditional differentiation of integration substrates between gyral backbone and sulcal circuits. Importantly, stronger differentiation does not necessarily indicate a better configuration, because extreme differentiations may indicate over-dependence on a single integration substrate. A stronger reliance on the gyral backbone (gyral-relative integration) may increase wiring cost and hub vulnerability. For instance, computational attack experiment showed that global efficiency may decrease disproportionately due to targeted disruption (Crossley et al., 2014). Dysfunctional hubs influenced the resilience of networks, related to cortical deficits or cognitive declines (Crossley et al., 2014; Deery et al., 2023). Conversely, excessive reliance on sulcal core-like profiles and local circuits (sulcal-relative integration) may reduce flexible long-range communication and alternative routing capacity (Kruggel & Solodkin, 2023). This interpretation is consistent with evidence that sulcal networks show relatively higher long-term test-retest reliability and significantly heritability, suggesting more stable but potentially less plastic organization (Lin et al., 2024). Thus, the optimal differentiation state may not be maximal gyral or sulcal integration, but an intermediate balance in which global and local processing are jointly supported. This was consistent with the continuous process of negotiating, and re-negotiating, trade-offs between physical cost and topological profiles (Bullmore & Sporns, 2012).

Additionally, the SI reflects the integrated coordination between differentiated substrates. An SI-integration state allowed locally sulcal components to be more effectively embedded into the gyral backbone through relatively high-cost and community-distributed bridge. This interpretation is consistent with the network integration principle, which characterized the potential for rapidly processing the specialized information across distributed systems (Rubinov & Sporns, 2010). These bridges allowed that gyral-sulcal substrates satisfied both local clustering and global interaction demands, supporting the functional segregation and integration with the economical reconfiguration (Bullmore & Sporns, 2012; Sporns & Honey, 2006). Such reconfigurations may support the diverse functional/cognitive demands and resilience (Cohen & D'Esposito, 2016; Rubinov & Sporns, 2010; Wang et al., 2021; Wig, 2017)

Epoch-dependent differentiation–coordination dynamics in gyral–sulcal networks

During development, the system showed a pattern consistent with constructive refinement. The trajectory moved toward an intermediate differentiation balance, accompanied by relatively strengthened gyral wiring and more integrated coordination with stronger wiring. This pattern may be driven by consolidation the gyral backbone integration and improved the gyral-sulcal coordination, shown in Fig. S5. One possible explanation is that, during development, the network architecture becomes more long-range distributed rather than more local (Fair et al., 2009), resulting in increased integration, decreased segregation and promoted global efficiency (Hagmann et al., 2010). Further optimization of network efficiency may be supported by the refinement of myelination and synaptic pruning, particularly in short-range fibers (Ouyang et al., 2017). Since costly structural hubs maintain a relatively consistent adult-like spatial distribution after their establishment (Cao et al., 2017; Grayson et al., 2014; Oldham & Fornito, 2019), the topological optimization may rely less on relocating hub architecture but more on refining the low-cost wiring allocation. In this state, improved spatial embedding of sulcal circuits may allow them to participate more efficiently in the gyral backbone, thereby supporting a hierarchically distributed integration architecture and promoting the reallocation of integration roles between gyral and sulcal substrates. Thus, differentiation shifts toward a slight population-biased sulcal-relative integration balance as maturation proceeds.

Young adulthood extends the developmental trajectory, but with weaker and more selective changes, indicating a late maturation or fine-tuning stage. In particular, G–S bridge pathways may become more refined, shown in Fig. S5, allowing the gyral backbone and sulcal circuits to maintain efficient coordination. This fine-tuning is consistent with evidence that local efficiency, rich-club organization, connection strength, and connection distance peak during young to middle adulthood (Cao et al., 2014). Additional evidence suggests that structural topology represent both increased global integration and local segregation until 32 years old (Mousley et al., 2025).

Midlife represented a key transition window. In this stage, fluctuations or turning patterns across the four DI/SI axes suggest that the mature gyral–sulcal architecture begins to renegotiate the balance between gyral backbone, sulcal circuits, and cross-type coordination. Mechanistically, this transition may be driven by reduced G–S weight, increased G–G and S–S connection length, and increasing segregation within these subnetworks, shown in Fig. S5. Such changes are consistent with the idea that costly long-range connections begin to lose stability while local circuits start to reorganize (Cao et al., 2014; Ma et al., 2021; Yeatman et al., 2014). For instance, the rich club architecture peaks at ~40 years (Yeatman et al., 2014) and the local efficiency is identified as the main predictor of midlife ages (Mousley et al., 2025). More broadly, recent reviews emphasize that midlife involves multiple cell-, tissue-, and organ -specific subprocesses, showing non-linearly abrupt changes, including turning, plateaus, breakpoints, and cyclicity loss (Dohm-Hansen et al., 2024).

Compared with development, aging showed opposing trends in cortical network reorganizations: the system moved away intermediate balance of integration substrates and toward more segregated cross-type coupling. Specifically, we observed weakening of the gyral backbone, consistent with the vulnerability of integrated hubs during aging (Coelho et al., 2021; Li et al., 2023). For instance, costly hubs with longer connections and higher metabolic rates are more susceptible to age-related insults than other regions, and are therefore particularly vulnerable to aging (Li et al., 2023). In contrast, due to spatial penalty and compensation, sulcal circuits may show local reinforcements. Specifically, generative modeling has suggested that aging brains favor short-range connections with low wiring cost over long-range connections under spatial penalty constraints (Zuo et al., 2017). In addition, compensation theories propose that some regions may show higher utility of neural resources to compensate deficits in other regions (Deery et al., 2023). Therefore, as long-range gyral integration declines, the system may rely more heavily on local sulcal circuits to maintain communication, consistently with increased weight and length of S–S/G–S connections, shown in Fig. S5. This may explain the deviation from the intermediate balance and the shift toward more sulcal relatively integration. However, local reinforcement may not preserve global integration, as we observed increased segregation and decreased integration across G–G, G–S, and S–S subnetworks, shown in Fig. S5. The cross-type synergy continued shifting toward more locally segregated reorganizations. Thus, aging may reflect compensatory rebalancing under declining wiring resources of folding organizations (Coelho et al., 2021; Deery et al., 2023; Mousley et al., 2025; Zuo et al., 2017).

Differentiation-supported youth and coordination-supported aging in executive function

EF may be jointly shaped by gyral–sulcal differentiation and cross-type coordination, rather than by stronger integration or segregation alone. Core EFs require coordination of multiple subnetworks, associated with both local specialization and globally integrative long-range connections (Baum et al., 2017; Cohen & D'Esposito, 2016; Crossley et al., 2014; Diamond, 2013). Within our folding-informed framework, we observed that EF appeared to be more associated with DI-related wiring differentiation during development. Specifically, gyral-sulcal maturation strengthens the gyral backbone while deepening the embedding of sulcal circuits, forming a hierarchically distributed integration pattern, support both local specialization and efficient global communication. This is consistent with prior structural network studies showing that both greater global efficiency and modular segregation in structural networks support executive function improvement during development (Baum et al., 2017). In midlife and aging, EF became increasingly dependent on preserved cross-type wiring coordination. This is consistent with studies showing that EF related to wide coordination across cortical and subcortical substructures, more vulnerable to network efficiency decline (Fjell et al., 2016). In addition, structural efficiency degeneration partially explained the reductions in executive function during aging (Fjell et al., 2016; King et al., 2018; Pedersen et al., 2021). Our semi_LM results further localized the EF-coupling transition: DI-FA2 contributed positively to EF-aligned changes during development, whereas SI-FA2 declined from midlife to aging despite remaining positively coupled with EF.

Limitations and future considerations:

Several limitations should be considered. First, this study used 1687 participants from HCP-D, HCP-YA and HCP-A datasets, covering ages 8–89 years. Both sample size and age coverage remain limited. Future work should incorporate more subjects from multi-cohort datasets with more wider age coverage. Second, the current analyses were based on cross-sectional data. Therefore, the observed age-related DI/SI trajectories did not reflect direct within-individual change. Longitudinal cohorts will be needed to test these trajectories within individuals. Third, the DI/SI framework depends on the definition of gyral and sulcal parcels and thresholding strategies. Although the current framework characterized gyral–sulcal differentiation and coordination, future studies should test whether the findings are robust across alternative gyral–sulcal definitions and different connectivity thresholds. Finally, the latent DI/SI axes were derived using data-driven factor analysis. These axes may vary with sample composition, feature scaling, and the set of topological measures included. Future work should validate the latent dimensions in independent datasets.

Conclusion:

In this study, we used structural connectivity from the HCP lifespan datasets to investigate how gyral–sulcal structural organization and its relationship with executive function changes from 8 to 89 years. We defined two complementary measures, the Differentiation Index (DI) and Synergy Index (SI), to quantify how gyri and sulci differ in their network organization and how they coordinate through cross-type connections. We found that DI and SI features form interpretable low-dimensional latent axes reflecting integration–segregation balance and wiring allocation. Lifespan trajectories projected into these latent axes revealed stage-specific gyral–sulcal reconfiguration of differentiation and coordination. Finally, differentiation-related wiring cost contributed to EF-aligned improvement during development and coordination-related wiring cost became increasingly relevant in later adulthood. Together, these findings suggest that gyral–sulcal differentiation and coordination provide a useful framework for understanding lifespan cortical reconfiguration and its age-dependent relationship with cognition.

Supplementary files

Fig. S1. Lifespan trajectories of absolute G–G, S–S, and G–S structural connectivity features. The x-axis represents age, and the y-axis represents the absolute feature value of gyral–gyral (G–G), sulcal–sulcal (S–S), and gyral–sulcal (G–S) connections. Dots and line styles follow Fig. 2.

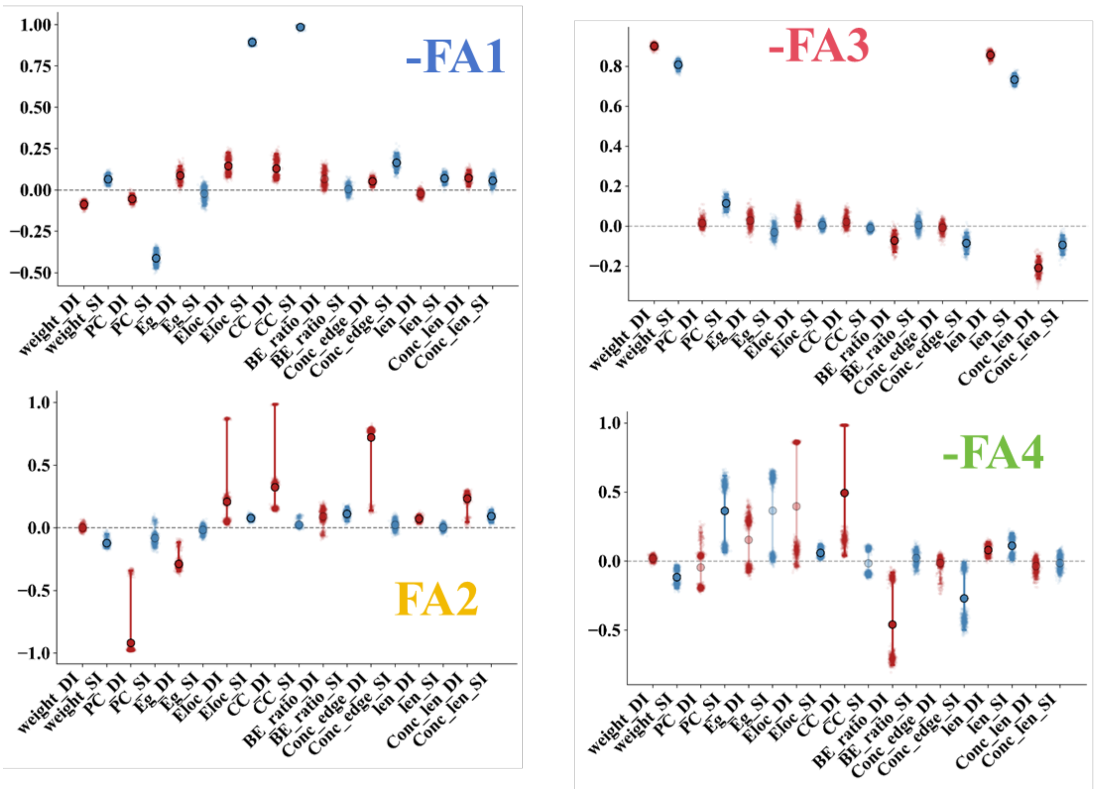

Fig. S2. Joint factor analysis was performed using both differentiation index (DI) and synergy index (SI) features to identify shared low-dimensional latent axes of gyral–sulcal structural organization. Factor 1 mainly corresponded to the SI-related integration–segregation axis, consistent with SI-FA1 in the separate FA model. Factor 2 primarily captured a DI-FA1 topology axis. Factor 3 reflected a shared strength/length scale across DI and SI features, whereas Factor 4 represented a distributed versus hub-based integration axis.

Epoch-wise feature–age association and driver analysis

To better interpret the age-related patterns, we performed additional subject-level analyses within 4 predefined lifespan epochs (development, young adulthood, midlife and aging). These analyses were based on the covariate-adjusted values derived from the GAMLSS models. For each epoch, we used LASSO regression to identify the significant topological multivariate. Age was used as the response variable, and the DI/SI features were used as predictors. In parallel, we computed Pearson correlations with age. Within each epoch, whereas LASSO was used to identify multivariate drivers, Pearson correlation was used to characterize the age-related change direction. These analyses provided an interpretable epoch-wise summary of DI/SI organizations.

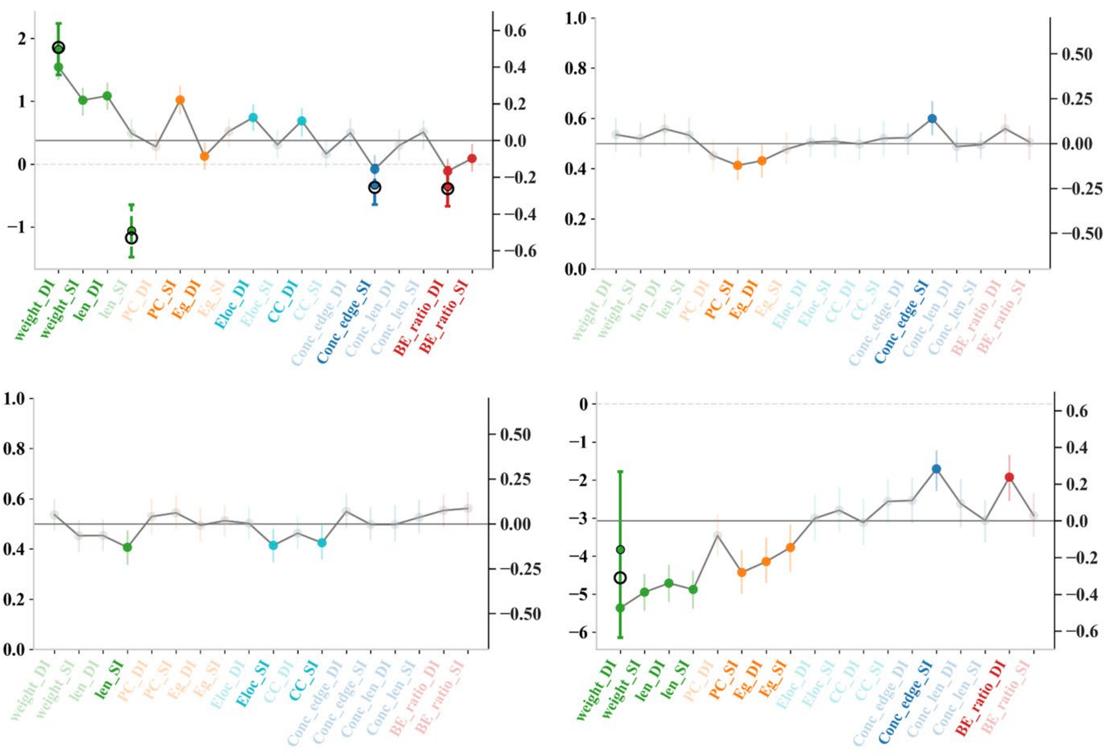

Fig. S3. Epoch-wise subject-level DI and SI organizations across four lifespan stages. The left y-axis represents the LASSO regression coefficient. Open black circles indicate the original coefficients estimated from the observed data, whereas filled points indicate the bootstrap mean coefficients. Vertical error bars represent bootstrap confidence intervals; coefficients whose confidence intervals included zero are not displayed. Solid error bars denote DI features, and dashed error bars denote SI features. In parallel, Pearson correlations between each feature and age were computed to characterize the direction of age-related change. The right y-axis represents the Pearson correlation coefficient (r). Non-significant correlations are shown in gray.

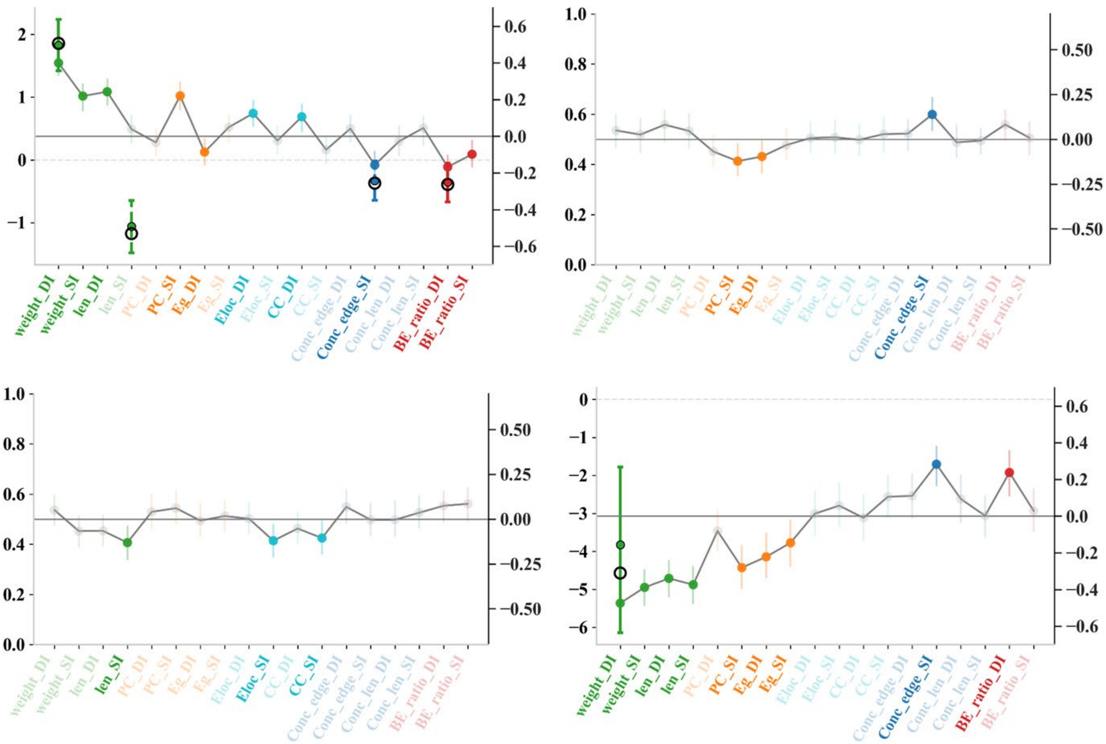

Fig. S4. Epoch-wise subject-level analyses of absolute G–G, G–S, and S–S structural features across the lifespan. The styles are similar to Fig. S3

Feature	Development (8–21.99)					Young adult (21.99–35.01)					Midlife (35.01–65.01)					Aging (65.01–89)				
	DI	SI	GG	SS	GS	DI	SI	GG	SS	GS	DI	SI	GG	SS	GS	DI	SI	GG	SS	GS
weight	+	+	+	–	–					–					–	–	–	–	+	+
len (percentile)	+			–	–							–	+	+		–	–		+	+
PC (integration)		+	+	+	+		–										–			–
Eg (integration)	–		+	+	+											–	–	–		–
Eloc (segregation)	+		–	–	–							–	+	+	+			+	+	+
CC (segregation)	+			–	–							–	+	+	+			+	+	+
Conc_edge (Commuities Concentration)		–			–		+			+							+			+
Conc_len (Commuities Concentration length)																				
BE_ratio (reliance on gyri or sulci)	–	–	–		–									–		+			–	

Fig. S5. Summary of significant age-related increase and decrease patterns across four lifespan epochs. The figure summarizes the epoch-wise results shown in Figs. S3 and S4 for DI, SI, and absolute GG, SS, and GS feature sets across development, young adulthood, midlife, and aging. A "+" indicates a significant positive association with age, and a "–" indicates a significant negative association with age. Blank cells indicate non-significant results. DI, Differentiation Index; SI, Synergy Index; GG, gyral–gyral; SS, sulcal–sulcal; GS, gyral–sulcal.

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